

Assignment 2

Due Monday 02/09/26

Reading:

- Required: Course Notes,
- Suggested: Ch 2-3 [EIT]

Problems:

1. (EIT: 2.04) *Entropy of functions of a random variable.* Let X be a discrete random variable. Show that the entropy of a function of X is less than or equal to the entropy of X by justifying the following steps:

$$\begin{aligned} H(X, g(X)) &\stackrel{(a)}{=} H(X) + H(g(X) | X) \\ &\stackrel{(b)}{=} H(X); \\ H(X, g(X)) &\stackrel{(c)}{=} H(g(X)) + H(X | g(X)) \\ &\stackrel{(d)}{\geq} H(g(X)). \end{aligned}$$

Thus $H(g(X)) \leq H(X)$.

2. (EIT: 2.12) *Example of joint entropy.* Let $p(x, y)$ be given by

X/Y	0	1
0	1/3	1/3
1	0	1/3

Find:

- (a) $H(X), H(Y)$.
 - (b) $H(X|Y), H(Y|X)$.
 - (c) $H(X, Y)$.
 - (d) $H(Y) - H(Y|X)$.
 - (e) $I(X; Y)$.
 - (f) Draw a Venn diagram for the quantities in parts (a) through (e).
3. (GRIT: 1.06) *Entropy of time to first success.* A fair coin is flipped repeatedly. Let X_i denote the number of flips required for the i th occurrence of a head. (i.e., if the coin is a head on the first flip then $X_1 = 1$). For example, the flip sequence “*TTTHTHHTTH*...” generates the outcome $(X_1, X_2, X_3, \dots) = (4, 2, 3, \dots)$.

- (a) Find the entropy $H(X_1)$ in bits.
 - (b) Find the conditional entropy $H(X_2|X_1)$ in bits.
 - (c) Show that $H(X_1) \leq H(X_2) \leq 2H(X_1)$. [Hint, try expressing $H(X_1, X_2)$ two different ways.]
4. (GRIT: 2.01) *Cross-Entropy Loss*. Consider a problem where we are given n iid samples $(x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$ drawn from $p(x, y)$ and the goal is to learn a predictor f that maps each $x \in \mathcal{X}$ to a prediction of $y \in \mathcal{Y}$. Without loss of generality, we will assume the prediction is an estimate of the posterior distribution of y given x . A common approach in machine learning is to define a loss function and then use the predictor that minimizes the average loss.

Let $\mathcal{P}(\mathcal{Y})$ denote the set of pmfs over $\mathcal{Y}$. Then, the cross-entropy loss $\ell: \mathcal{Y} \times \mathcal{P}(\mathcal{Y}) \rightarrow \mathbb{R}_{\geq 0}$ maps the true $y \in \mathcal{Y}$ and the predicted posterior $q(y)$ to the non-negative real loss

$$\ell(y, q) = \log \frac{1}{q(y)}.$$

For each $\theta \in \mathbb{R}^d$, let $f_\theta(x): \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$ be a candidate predictor that maps each $x \in \mathcal{X}$ to a pmf over $\mathcal{Y}$. The, the average cross-entropy loss is given by

$$L_n(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_\theta(x_i)).$$

The following should help illuminate an important connection between machine learning and information theory. Note: One can denote the pmf $f_\theta(x)$ evaluated at $y \in \mathcal{Y}$ by $f_\theta(x)(y)$.

- (a) To what non-random value will $L_n(\theta)$ converge to as $n \rightarrow \infty$?
 - (b) For fixed θ , write the quantity in (a) as $H(Y|X)$ plus the average (over $\mathcal{X}$) divergence between $p(y|x)$ and $q(y|x) = f_\theta(x)(y)$.
 - (c) Use the non-negativity of divergence to identify the minimum possible cross-entropy loss and choice of f_θ that achieves it.
5. (EIT: 3.04) *AEP*. Let X_i be iid $\sim p(x)$, $x \in \{1, 2, \dots, m\}$. Let $\mu = \mathbb{E}[X]$ and $H = H(X)$. Let

$$A^n = \{x^n \in \mathcal{X}^n : |-\frac{1}{n} \log p(x^n) - H| \leq \epsilon\}$$

$$B^n = \{x^n \in \mathcal{X}^n : |\frac{1}{n} \sum_{i=1}^n X_i - \mu| \leq \epsilon\}$$

- (a) Does $\mathbb{P}[X^n \in A^n] \rightarrow 1$?
- (b) Does $\mathbb{P}[X^n \in A^n \cap B^n] \rightarrow 1$?
- (c) Show that $|A^n \cap B^n| \leq 2^{n(H+\epsilon)}$
- (d) Show that $|A^n \cap B^n| \geq (\frac{1}{2})2^{n(H-\epsilon)}$ for n large enough

6. (EIT: 2.29) *Inequalities.* Let X , Y and Z be joint random variables. Prove the following inequalities and find conditions for equality.

- (a) $H(X, Y|Z) \geq H(X|Z)$.
- (b) $I(X, Y; Z) \geq I(X; Z)$.
- (c) $H(X, Y, Z) - H(X, Y) \leq H(X, Z) - H(X)$.
- (d) $I(X; Z|Y) \geq I(Z; Y|X) - I(Z; Y) + I(X; Z)$.

Practice Problems (do not hand in):

1. (GRIT: 1.01) *Union Bounds.* Let $A_1, A_2, \dots$, be events defined on a sample space Ω . Prove the following two inequalities.

- (a) $\mathbb{P}\left[\bigcup_{i=1}^{\infty} A_i\right] \leq \sum_{i=1}^{\infty} \mathbb{P}[A_i]$
- (b) $\mathbb{P}\left[\bigcap_{i=1}^{\infty} A_i\right] \geq 1 - \sum_{i=1}^{\infty} \mathbb{P}[A_i^C]$

2. (GRIT: 1.02) *Independence.* Construct examples that meet the following conditions.

- (a) Three discrete random variables X , Y , Z that are not independent but are pairwise independent.
- (b) Three discrete random variables X , Y , and Z such that X and Y are dependent but X and Y are conditionally independent given Z .

3. (EIT: 2.01) *Coin flips.* A fair coin is flipped until the first head occurs. Let X denote the number of flips required.

- (a) Find the entropy $H(X)$ in bits. The following formulas may be useful:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, \quad \sum_{n=0}^{\infty} nr^n = \frac{r}{(1-r)^2}.$$

- (b) A random variable X is drawn according to this distribution. Find an “efficient” sequence of yes-no questions of the form: “Is X contained in the set S ”. Compare $H(X)$ to the expected number of questions required to determine X .

4. (EIT: 3.01) *Markov’s inequality and Chebyshev’s inequality*

- (a) (*Markov’s inequality*) For any nonnegative random variable X and any $t > 0$, show that

$$\Pr\{X \geq t\} \leq \frac{EX}{t}.$$

Exhibit a random variable that achieves this inequality with equality.

- (b) (*Chebyshev's inequality*) Let Y be a random variable with mean μ and variance σ^2 . By letting $X = (Y - \mu)^2$, show that for any $\epsilon > 0$

$$\Pr\{|Y - \mu| > \epsilon\} \leq \frac{\sigma^2}{\epsilon^2}.$$

- (c) (*Weak law of large numbers*) Let $Z_1, Z_2, \dots, Z_n$ be a sequence of i.i.d. random variables with mean μ and variance σ^2 . Let $\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i$ be the sample mean. Show that

$$\Pr\{|\bar{Z}_n - \mu| > \epsilon\} \leq \frac{\sigma^2}{n\epsilon^2}.$$

Thus, $\Pr\{|\bar{Z}_n - \mu| > \epsilon\} \rightarrow 0$ as $n \rightarrow \infty$. This is known as the *weak law of large numbers*.

5. (EIT: 2.32) *Fano*. We are given the following joint distribution on (X, Y) :

X/Y	a	b	c
1	1/6	1/12	1/12
2	1/12	1/6	1/12
3	1/12	1/12	1/6

Let $\hat{X}(Y)$ be an estimator for X (based on Y) and let $P_e = \Pr(\hat{X}(Y) \neq X)$.

- (a) Find the minimum probability of error estimator $\hat{X}(Y)$ and the associated P_e .
(b) Evaluate Fano's inequality for this problem and compare.
6. (EIT: 2.28) *Mixing increases entropy*. Show that the entropy of the probability distribution, $(p_1, \dots, p_i, \dots, p_j, \dots, p_m)$, is less than the entropy of the distribution $(p_1, \dots, \frac{p_i + p_j}{2}, \dots, \frac{p_i + p_j}{2}, \dots, p_m)$. Show that in general any transfer of probability that makes the distribution more uniform increases the entropy.